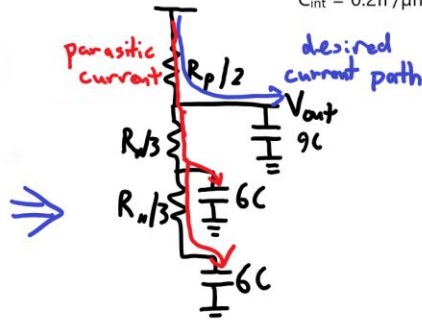
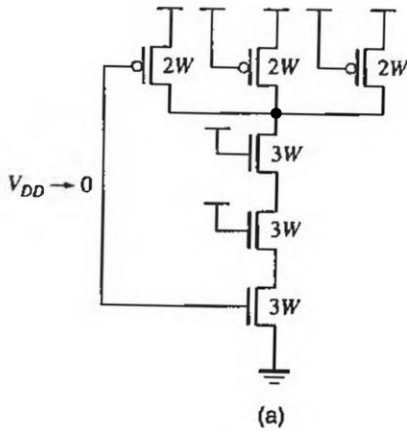


U

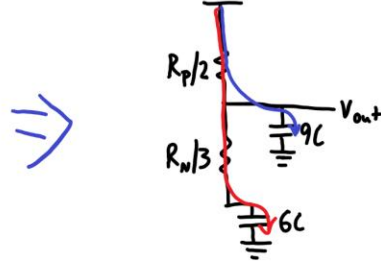
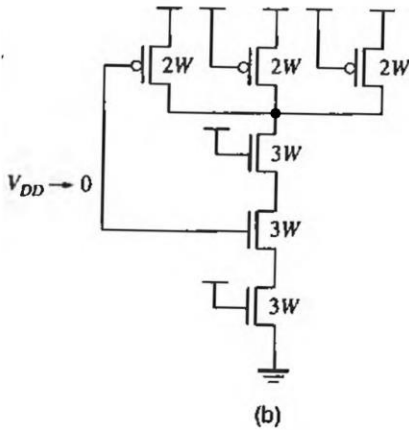
For 130nm process, $W = 1 = 100\text{nm}$
 $C_{\text{self}} = 1\text{fF}/\mu\text{m} = 0.1\text{fF}/100\text{nm}$
 $C_g = 2\text{fF}/\mu\text{m} = 0.2\text{fF}/100\text{nm}$
 $C_{\text{int}} = 0.2\text{fF}/\mu\text{m}$ of wire length



$$\tau_a = R_p/2(9C) + R_p/2(12C)$$

$$\tau_a = 10.5 R_p C = 10.5 \cdot 30\text{k} \cdot 0.1\text{f}$$

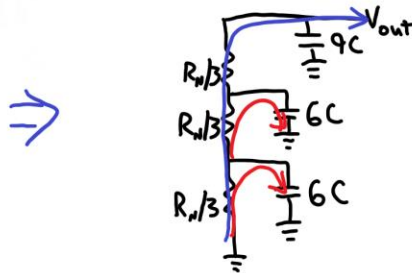
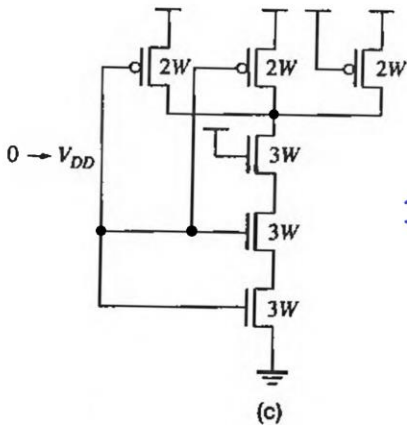
$$\tau_a = 31.5\text{ps}$$



$$\tau_b = R_p/2(9C) + R_p/2(6C)$$

$$\tau_b = 7.5 R_p C = 7.5 \cdot 30\text{k} \cdot 0.1\text{f}$$

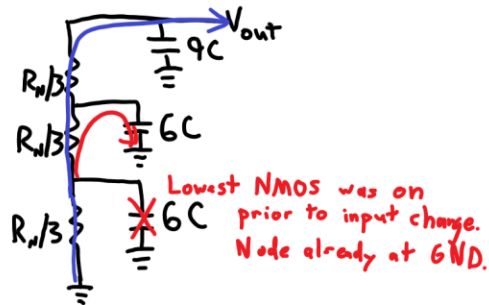
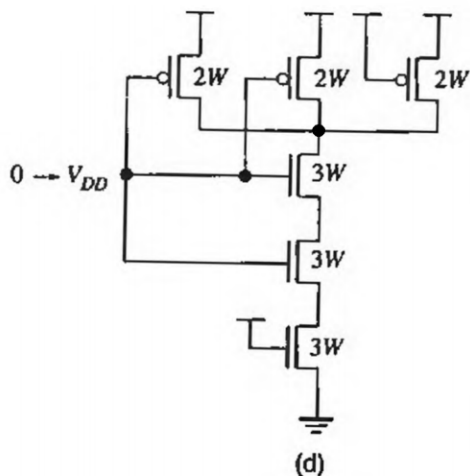
$$\tau_b = 22.5\text{ps}$$



$$\tau_c = \frac{R_N}{3} \cdot 6C + 2 \frac{R_N}{3} \cdot 6C + 3 \frac{R_N}{3} \cdot 9C$$

$$\tau_c = 15 R_N C = 15 \cdot 12.5\text{k} \cdot 0.1\text{f}$$

$$\tau_c = 18.75\text{ps}$$



$$\tau_d = 2 \frac{R_n}{3} \cdot 6C + 3 \frac{R_n}{3} \cdot 9C$$

$$\tau_d = 13 R_n C = 13 \cdot 12.5k \cdot 0.1f$$

$$\tau_d = 16.25 \text{ ps}$$

ii) $C_g = (W_p + W_n) \cdot 0.2f = (3) \cdot 0.2f = 0.6fF$

a) $\tau_a = 31.5ps + \frac{R_p}{2} C_g = 31.5ps + \frac{30k}{2} \cdot 0.6f = 40.5ps$

b) $\tau_b = 22.5ps + \frac{R_p}{2} C_g = 22.5ps + 9p = 31.5ps$

c) $\tau_c = 18.75ps + \frac{3R_n}{3} C_g = 18.75ps + 12.5k \cdot 0.6f = 26.25ps$

d) $\tau_d = 16.25ps + \frac{3R_n}{3} C_g = 16.25ps + 7.5p = 23.75ps$

iii)

$$C_{int} = 0.2f \cdot 100 = 20fF$$

Like in part ii, additional capacitance is multiplied by R_{eff} at output.

For a and b, $R_{eff} = \frac{R_p}{2} = \frac{30k}{2} = 15k\Omega$

For c and d, $R_{eff} = \frac{3R_n}{3} = 12.5k\Omega$

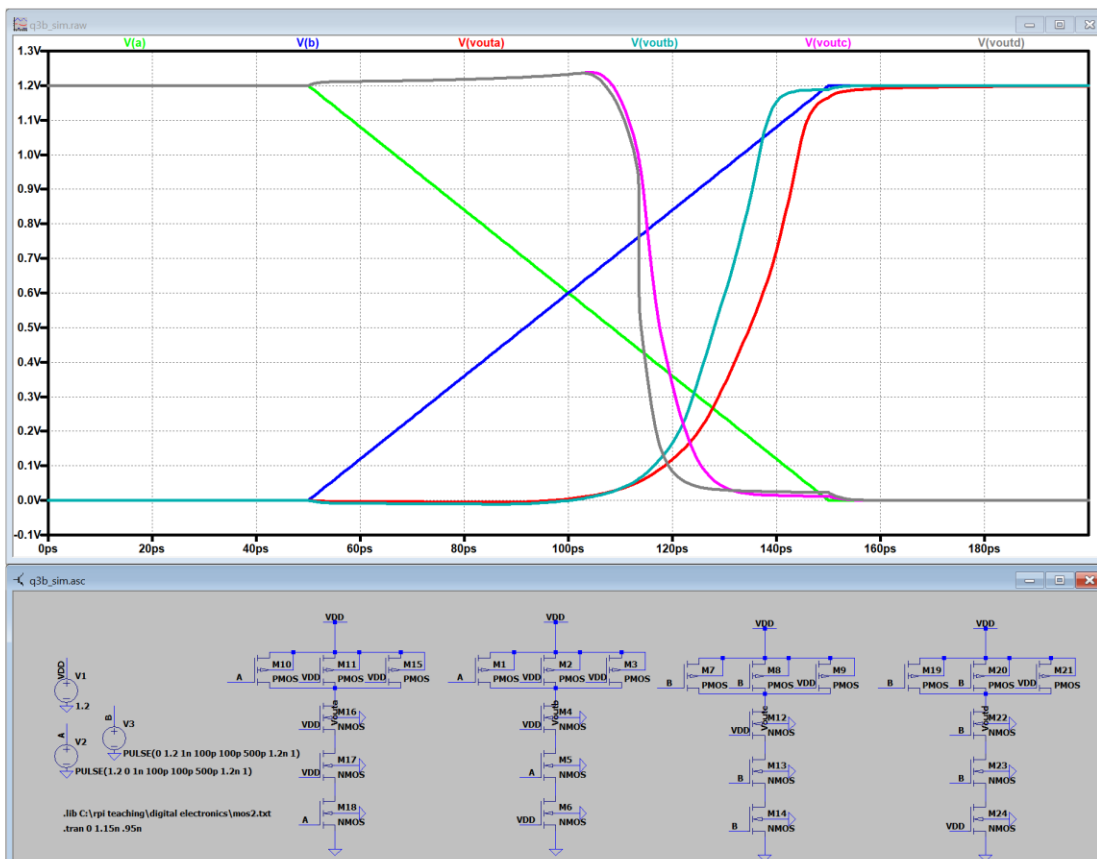
a) $\tau_a = 40.5ps + 15k \cdot 20f = 40.5ps + 300p = 340.5ps$

b) $\tau_b = 31.5ps + 15k \cdot 20f = 331.5ps$

c) $\tau_c = 26.25ps + 12.5k \cdot 20f = 26.25ps + 250p = 276.25ps$

d) $\tau_d = 23.75ps + 12.5k \cdot 20f = 23.75ps + 250p = 273.75ps$

1b)



By Inspection from LTspice:

$$\begin{aligned}\tau_a &= 37.6\text{ps} \\ \tau_b &= 30.2\text{ps} \\ \tau_c &= 16.7\text{ps} \\ \tau_d &= 13.5\text{ps}\end{aligned}$$

LTspice results have slightly shorter fall time, and slightly longer rise time delays than hand calculations. Despite these differences, the simulation results follow the same delay trend as predicted by Elmore delay hand calculations.

2a)

$$\Delta V_{\text{victim}} = \frac{C_c \Delta V_{\text{aggressor}}}{(C_{\text{gnd}} + C_c)}$$

i. $\Delta V_{\text{victim}} = \frac{100\text{f} \cdot (1.2 - 0)}{60\text{f} + 100\text{f}}$

$\Delta V_{\text{victim}} = 0.75\text{V}$

ii. $\Delta V_{\text{victim}} = \frac{100\text{f} \cdot (0 - 1.2)}{60\text{f} + 100\text{f}}$

$\Delta V_{\text{victim}} = -0.75\text{V}$

2b)

$$C_L = C_{\text{gnd}} + 2C_c \quad \text{both switching opposite}$$

$$C_L = 60\text{f} + 2 \cdot 100\text{f}$$

$$C_L = 260\text{fF}$$

i. $\frac{\tau_{\text{opposite}}}{\tau_{\text{same}}} = \frac{R C_{\text{opp}}}{R C_{\text{same}}} = \frac{260\text{f}}{60\text{f}} = 4.33$

333% increase